PHYDSC403T — Basic Quantum Mechanics

Unit 1: Schrödinger Equation and Operators

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC403T, Full Marks 100 [ESE 70 + CCA 30] | Unit 1: 10 lectures

Notes 1 of 20 — written from published university lecture notes (Indian universities and colleges first, then IIT / NPTEL, then IGNOU open-learning material). Full list of sources at the end of these notes.

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Unit syllabus (as in the syllabus file)

1. Probability and probability current density.
2. Equation of continuity and conservation of probability.
3. Quantum mechanical operators and their algebra, linear operator, commutation relations.
4. Simultaneous eigenfunction. Compatible and incompatible observables.
5. Eigenvalue and eigenfunction, position operator, Hamiltonian operator, kinetic energy operator, linear momentum operator.
6. Hermitian operators and their properties.

Top 2 high-mark topics in this unit

1. Hermitian operators and their properties, with commutation relations and compatible / incompatible observables — AU already asks a commutator / Hermitian question, and operator algebra is the derivation core of the unit.
2. Probability current density and equation of continuity / conservation of probability — named first in the syllabus, derivation-heavy, and an AU gap, so high risk as a long question.

Contents

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1. The wave function, probability density and well-behaved wave functions

The state of a particle is described by a wave function $\psi(\mathbf{r}, t)$, in general complex. It contains all the information that can be known about the particle.

Born interpretation. $|\psi(\mathbf{r}, t)|^2$ is the probability density: the probability of finding the particle in a small volume $d\tau$ around $\mathbf{r}$ at time t is

$$dP = |\psi(\mathbf{r}, t)|^2 \, d\tau = \psi^* \psi \, d\tau$$

So the probability density is $\rho(\mathbf{r}, t) = \psi^* \psi = |\psi|^2 \geq 0$. The square of the wave function is a measure of the probability of finding a particle at a certain point in space.

Normalisation. The particle must be somewhere, so

$$\int_{\text{all space}} |\psi|^2 \, d\tau = 1$$

A wave function that satisfies this is called normalised. If $\int |\psi|^2 d\tau = N$, the normalised function is $\psi / \sqrt{N}$.

Well-behaved wave function. A physically acceptable ψ must be finite everywhere (it cannot blow up to infinity), single-valued (one value of ψ at each point), continuous, and its first derivative $d\psi/dx$ continuous, except where the potential is infinite, and square-integrable, i.e. $\int |\psi|^2 d\tau$ is finite, so that it can be normalised. A wave function with these attributes is called a well-behaved wave function.

Linear combination. If ψ_1 and ψ_2 are well-behaved, then $c_1\psi_1 + c_2\psi_2$ is also well-behaved: a sum of finite, single-valued, continuous functions is finite, single-valued and continuous, and it is square-integrable. This is the superposition principle in its simplest form. (AU 2023, Sec A.)

2. Schrödinger equation

Time-dependent Schrödinger equation (TDSE). For a particle of mass m in a potential $V(\mathbf{r}, t)$,

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t) \right] \psi$$

where $\hbar = h/2\pi$ and $\hat{H}$ is the Hamiltonian operator, the sum of the kinetic energy and the potential energy operators.

Time-independent Schrödinger equation (TISE). If V does not depend on time, write $\psi(\mathbf{r}, t) = \phi(\mathbf{r}) e^{-iEt/\hbar}$. Then ϕ satisfies

$$\hat{H}\phi = E\phi, \qquad -\frac{\hbar^2}{2m}\nabla^2\phi + V(\mathbf{r})\phi = E\phi$$

Such solutions are **stationary states**: their probability density $|\psi|^2 = |\phi|^2$ does not change with time, although ψ itself has the phase factor $e^{-iEt/\hbar}$. The eigenvalue equation $\hat{H}\psi = E\psi$ is referred to as the time-independent Schrödinger equation.

3. Probability current density, continuity equation and conservation of probability

The way to derive the probability current is to start from the time-dependent Schrödinger equation and its complex conjugate. The time-dependent equation is

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi$$

and its complex conjugate, remembering that the potential V is a real function, is

$$-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi^* + V\psi^*$$

Multiply the first equation by ψ^* and the second by ψ, and subtract one from the other. The potential term cancels out on both sides when we subtract, no matter what the potential is, and we get

$$i\hbar\left(\psi^*\frac{\partial\psi}{\partial t}+\psi\frac{\partial\psi^*}{\partial t}\right)=-\frac{\hbar^2}{2m}\left(\psi^*\nabla^2\psi-\psi\nabla^2\psi^*\right)$$

The left hand side is $i\hbar$ times the time derivative of $\psi^*\psi$, that is, of the probability density ρ:

$$i\hbar\,\frac{\partial\rho}{\partial t}=i\hbar\,\frac{\partial(\psi^*\psi)}{\partial t}$$

The right hand side looks like the divergence of a current. To see this, use the formula for the divergence of a scalar times a vector, $\nabla\cdot(u\mathbf{V})=\nabla u\cdot\mathbf{V}+u\,\nabla\cdot\mathbf{V}$. Applying it to the two terms,

$$\nabla\cdot(\psi^*\nabla\psi)=\nabla\psi^*\cdot\nabla\psi+\psi^*\nabla^2\psi$$

$$\nabla\cdot(\psi\nabla\psi^*)=\nabla\psi\cdot\nabla\psi^*+\psi\nabla^2\psi^*$$

Subtracting the second from the first, the cross terms $\nabla\psi^*\cdot\nabla\psi$ and $\nabla\psi\cdot\nabla\psi^*$ are the same, so they cancel out, and what is left is exactly the combination on the right hand side:

$$\psi^*\nabla^2\psi-\psi\nabla^2\psi^*=\nabla\cdot(\psi^*\nabla\psi-\psi\nabla\psi^*)$$

That is the step that turns the right hand side into a divergence. Dividing both sides by $i\hbar$, we get the equation of continuity,

$$\frac{\partial\rho}{\partial t}+\nabla\cdot\mathbf{J}=0$$

where the probability current density $\mathbf{J}$ is

$$\mathbf{J}=\frac{\hbar}{2mi}\left(\psi^*\nabla\psi-\psi\nabla\psi^*\right)=\frac{\hbar}{m}\,\mathrm{Im}\left(\psi^*\nabla\psi\right)$$

In one dimension, $J_x=\dfrac{\hbar}{2mi}\left(\psi^*\dfrac{d\psi}{dx}-\psi\dfrac{d\psi^*}{dx}\right)$. The current is a real quantity: it is a complex number minus its complex conjugate, divided by $2i$. For a free-particle plane wave $\psi=Ae^{i(kx-\omega t)}$, $\rho=|A|^2$ and $J_x=\dfrac{\hbar k}{m}|A|^2=v\rho$, with $v=\hbar k/m$: probability flows with the particle velocity, as expected. If ψ is real (up to a constant phase), then $\psi^*\nabla\psi-\psi\nabla\psi^*=0$, so $\mathbf{J}=0$: no probability flows. This is true for bound stationary states with real ϕ.

Meaning. This is the equation of continuity, exactly like charge conservation in electrodynamics: probability is not created or destroyed, it only flows. Where ρ decreases, $\mathbf{J}$ carries probability away, and vice versa.

Conservation of probability. Integrate the continuity equation over all space. By the divergence theorem $\int\nabla\cdot\mathbf{J}\,d\tau$ becomes a surface integral at infinity, where $\psi\to0$, so it vanishes:

$$\frac{d}{dt}\int|\psi|^2\,d\tau=0$$

Total probability stays equal to 1 for all time. A wave function normalised at $t=0$ stays normalised, since total probability is conserved always.

4. Operators, linear operators and operator algebra

An **operator** is a rule that turns one function into another function: $\hat{A}\psi=\phi$. In quantum mechanics every observable is represented by an operator, which operates upon the wave function of the system.

Linear operator. Most of the relevant operators in quantum mechanics have the property of linearity. For any two wave functions ψ_1 and ψ_2, a linear operator $\hat{A}$ satisfies

$$\hat{A}(c_1\psi_1+c_2\psi_2)=c_1\hat{A}\psi_1+c_2\hat{A}\psi_2$$

where c_1 and c_2 are complex numbers. Differentiation d/dx, multiplication by x, and the Hamiltonian are all linear. For example, $\hat{x}(a\psi_1+b\psi_2)=a\hat{x}\psi_1+b\hat{x}\psi_2$, so $\hat{x}$ is a linear operator, and similarly $\hat{p}_x(a\psi_1+b\psi_2)=a\hat{p}_x\psi_1+b\hat{p}_x\psi_2$. All operators in quantum mechanics are linear.

Operator algebra. Operators can be added, $\hat{A}+\hat{B}$, and multiplied, $\hat{A}\hat{B}$ meaning "$\hat{B}$ first, then $\hat{A}$". In general the product of two operators does not obey the commutative law, i.e. $\hat{A}\hat{B}\neq\hat{B}\hat{A}$. The **commutator** measures the failure to commute:

$$[\hat{A},\hat{B}]=\hat{A}\hat{B}-\hat{B}\hat{A}$$

If the commutator of the operators $\hat{A}$ and $\hat{B}$ is zero, we say that $\hat{A}$ and $\hat{B}$ commute, i.e. $\hat{A}\hat{B}=\hat{B}\hat{A}$, and then we can interchange their order. The **anticommutator** is $\{\hat{A},\hat{B}\}=\hat{A}\hat{B}+\hat{B}\hat{A}$.

5. Commutation relations

The fundamental commutator. In one dimension, $\hat{x}=x$ (multiply by x) and $\hat{p}_x=-i\hbar\,d/dx$. To find out whether they commute, evaluate the action of the commutator $[\hat{x},\hat{p}_x]$ on an arbitrary wave function $\psi(x)$:

$$[\hat{x},\hat{p}_x]\psi=\hat{x}\hat{p}_x\psi-\hat{p}_x\hat{x}\psi$$

The first term is $\hat{x}\hat{p}_x\psi=x(-i\hbar\,d\psi/dx)=-i\hbar\,x\psi'$. The second term is $\hat{p}_x\hat{x}\psi=-i\hbar\,d(x\psi)/dx$, and by the product rule $d(x\psi)/dx=\psi+x\psi'$, so it becomes $-i\hbar\psi-i\hbar\,x\psi'$. Subtracting,

$$[\hat{x},\hat{p}_x]\psi=-i\hbar\,x\psi'-(-i\hbar\psi-i\hbar\,x\psi')=i\hbar\psi$$

Since the wave function ψ is arbitrary, we can write, as an operator identity,

$$[\hat{x},\hat{p}_x]=i\hbar$$

So the operators $\hat{x}$ and $\hat{p}_x$ do not commute with one another, and the value of the commutation bracket is $i\hbar$. In three dimensions,

$$[\hat{x}_i,\hat{p}_j]=i\hbar\,\delta_{ij},\qquad[\hat{x}_i,\hat{x}_j]=[\hat{p}_i,\hat{p}_j]=0$$

where $\delta_{ij}=1$ if $i=j$ and 0 otherwise. Position and momentum in *different* directions commute; in the *same* direction they do not. The commutator of two operators is important when the two operators correspond to canonically conjugate variables like x and p_x, y and p_y, z and p_z.

Rules of commutator algebra. These are easy to prove using the definition of a commutator. For operators $\hat{A}$, $\hat{B}$ and $\hat{C}$:

$$[\hat{A},\hat{B}]=-[\hat{B},\hat{A}]$$

$$[\hat{A},\hat{B}+\hat{C}]=[\hat{A},\hat{B}]+[\hat{A},\hat{C}]$$

$$[\hat{A},\hat{B}\hat{C}]=[\hat{A},\hat{B}]\hat{C}+\hat{B}[\hat{A},\hat{C}]$$

$$[\hat{A}\hat{B},\hat{C}]=\hat{A}[\hat{B},\hat{C}]+[\hat{A},\hat{C}]\hat{B}$$

$$[\hat{A},\hat{A}^n]=0,\qquad[\hat{A},c]=0\ \text{ for a constant } c$$

More commutators (learn to derive, not just memorise).

$$[\hat{x},\hat{p}_x^2]=2i\hbar\hat{p}_x$$

$$[\hat{p}_x,\hat{x}^2]=-2i\hbar\hat{x}$$

$$[\hat{x},f(\hat{p}_x)]=i\hbar\frac{\partial f}{\partial p_x}$$

$$[\hat{p}_x,f(x)]=-i\hbar f'(x)$$

For example, using the product rule for commutators, $[\hat{x},\hat{p}_x^2]=[\hat{x},\hat{p}_x]\hat{p}_x+\hat{p}_x[\hat{x},\hat{p}_x]=i\hbar\hat{p}_x+\hat{p}_x i\hbar=2i\hbar\hat{p}_x$.

Why commutators matter. They decide whether two observables can be known together (Section 7), and they give the uncertainty principle: for any two observables, $\Delta A\,\Delta B\geq\frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|$. For x and p_x this is $\Delta x\,\Delta p_x\geq\hbar/2$.

6. Eigenvalues, eigenfunctions and simultaneous eigenfunctions

Eigenvalue equation. If, for an operator $\hat{A}$, there is a (non-zero) function ψ_n and a number a_n such that

$$\hat{A}\psi_n = a_n\psi_n$$

then a_n is an **eigenvalue** of $\hat{A}$ and ψ_n is the corresponding **eigenfunction**. The set of all eigenvalues is the spectrum of $\hat{A}$. Measuring the observable A can only give one of its eigenvalues.

Degeneracy. If two or more independent eigenfunctions share the same eigenvalue, that eigenvalue is degenerate, and the number of such eigenfunctions is the degree of degeneracy.

Simultaneous eigenfunctions. Two operators $\hat{A}$ and $\hat{B}$ have a common, complete set of eigenfunctions if and only if they commute, $[\hat{A},\hat{B}]=0$. To see why, suppose $\hat{A}\psi = a\psi$ with a non-degenerate. Then $\hat{A}(\hat{B}\psi) = \hat{B}\hat{A}\psi = a(\hat{B}\psi)$, so $\hat{B}\psi$ is also an eigenfunction of $\hat{A}$ with eigenvalue a. By non-degeneracy $\hat{B}\psi$ must be a multiple of ψ itself, $\hat{B}\psi = b\psi$ for some number b: ψ is an eigenfunction of $\hat{B}$ too. If two operators commute, they possess a set of common eigenvectors. The argument extends to the degenerate case by choosing the right combinations inside each degenerate subspace.

7. Compatible and incompatible observables

A pair of observables A and B, corresponding to the operators $\hat{A}$ and $\hat{B}$, are **compatible** if the operators commute, $[\hat{A},\hat{B}]=0$, and **incompatible** if the operators do not commute, $[\hat{A},\hat{B}]\neq 0$. The reason for this classification becomes clear when we measure the two observables one after the other on the same system.

Suppose we first measure A and get the eigenvalue a_i; the state of the system immediately after the measurement is the eigenstate $|\phi_i\rangle$. Now measure B immediately after. If A and B are compatible, the operators commute and have a common set of eigenvectors, so $|\phi_i\rangle$ is also an eigenvector of $\hat{B}$, with some eigenvalue b_i. We get the value b_i, the system stays in the state $|\phi_i\rangle$, and if we measure A again we would certainly get a_i back. The second measurement does not destroy the information from the first, so it is possible to determine both A and B precisely at the same time. This holds whether the eigenvalues are degenerate or not.

If A and B are incompatible, they do not have simultaneous eigenvectors: $|\phi_i\rangle$ is not an eigenstate of $\hat{B}$. The measurement of B gives some eigenvalue b_j and leaves the system in the eigenstate of $\hat{B}$, $|\chi_j\rangle$. If we now measure A again, we could get any of the eigenvalues of $\hat{A}$, not just the one we got in the first measurement. So for incompatible observables, the second measurement scrambles the results of the first, and both cannot be measured sharply at the same time.

Example: $\hat{p}_x, \hat{p}_y, \hat{p}_z$ commute with each other; $\hat{H}$ and $\hat{p}_x$ commute for a free particle ($V=0$), so energy and momentum can be known together for a free particle. Position and momentum in the same direction do not commute, $[\hat{x},\hat{p}_x] = i\hbar \neq 0$; also $[\hat{L}_x,\hat{L}_y] = i\hbar\hat{L}_z$ for angular momentum. The product of their uncertainties obeys $\Delta A\,\Delta B \geq \tfrac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|$.

Pair	Commutator	Compatible?
$\hat{x}, \hat{p}_x$	$i\hbar$	No
$\hat{x}, \hat{p}_y$	0	Yes
$\hat{x}, \hat{H}$ (with $V=V(x)$)	$\neq 0$ in general	No
$\hat{p}_x, \hat{H}$ (free particle)	0	Yes

8. Position, momentum, kinetic energy and Hamiltonian operators

The position and momentum operators are used in the construction of operators of other dynamical variables such as energy. They are postulated, not proved: the position operator is $\hat{x} = x$ (multiplication by x), and the momentum operator is $\hat{p}_x = -i\hbar\, d/dx$. All operators in quantum mechanics can be constructed from the basic operators for position and momentum.

Observable	Operator (3D)	Operator (1D, x)
Position	$\hat{\mathbf{r}} = \mathbf{r}$ (multiply)	$\hat{x} = x$
Linear momentum	$\hat{\mathbf{p}} = -i\hbar\nabla$	$\hat{p}_x = -i\hbar\dfrac{d}{dx}$
Kinetic energy	$\hat{T} = \dfrac{\hat{\mathbf{p}}^2}{2m} = -\dfrac{\hbar^2}{2m}\nabla^2$	$\hat{T} = -\dfrac{\hbar^2}{2m}\dfrac{d^2}{dx^2}$
Potential energy	$\hat{V} = V(\mathbf{r})$ (multiply)	$\hat{V} = V(x)$

Total energy (Hamiltonian)	$\hat{H} = \hat{T} + \hat{V} = -\dfrac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})$	$\hat{H} = -\dfrac{\hbar^2}{2m}\dfrac{d^2}{dx^2} + V(x)$

Expectation value. The average value in quantum mechanics for any variable A in the state ψ is given by a postulate:

$$\langle A \rangle = \frac{\int \psi^* \hat{A} \psi \, d\tau}{\int \psi^* \psi \, d\tau} = \int \psi^* \hat{A} \psi \, d\tau \qquad (\psi \text{ normalised})$$

For example $\langle x \rangle = \int \psi^* x \psi \, dx$ and $\langle p_x \rangle = \int \psi^* \left(-i\hbar \dfrac{d\psi}{dx} \right) dx$.

9. Hermitian operators and their properties

Definition. An operator $\hat{A}$ is termed **Hermitian** (self-adjoint) if it satisfies $\hat{A}^\dagger = \hat{A}$, i.e. for all well-behaved ϕ, ψ (vanishing at infinity),

$$\int \phi^* (\hat{A}\psi) \, d\tau = \int (\hat{A}\phi)^* \psi \, d\tau, \qquad \text{in short } \langle \phi | \hat{A} | \psi \rangle = \langle \psi | \hat{A} | \phi \rangle^*$$

Observables must be represented by Hermitian operators, because measured values are real numbers. In this chapter we define Hermitian operators and show that they possess real eigenvalues.

Which operators are Hermitian? The position operator $\hat{x}$ is Hermitian, since $x = x^*$ for x real. The momentum operator $\hat{p}_x = -i\hbar \, d/dx$ is again Hermitian: by integration by parts, $\int \phi^* (-i\hbar \psi') dx = [-i\hbar \phi^* \psi] + \int (i\hbar \phi'^*) \psi dx$, and the boundary term vanishes for well-behaved functions, leaving $\int (-i\hbar \phi')^* \psi dx$. Note d/dx alone is *not* Hermitian; the factor $-i$ is what makes $\hat{p}_x$ Hermitian. The Hamiltonian $\hat{H} = \hat{T} + \hat{V}$ with real V is Hermitian. For Hermitian operators one writes $\hat{A}^\dagger = \hat{A}$, and moreover $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$.

Property 1 — eigenvalues are real. Let $\hat{A}$ be Hermitian with eigenfunction χ and eigenvalue λ, $\hat{A}\chi = \lambda\chi$. From the definition of a Hermitian operator, with $\phi = \psi = \chi$,

$$\int \chi^* \hat{A} \chi \, dx - \int \chi (\hat{A}\chi)^* \, dx = 0$$

Putting $\hat{A}\chi = \lambda\chi$ in both terms,

$$\lambda \int \chi^* \chi \, dx - \lambda^* \int \chi \chi^* \, dx = 0, \qquad \text{i.e.} \qquad (\lambda - \lambda^*) \int |\chi|^2 \, dx = 0$$

Since χ is a well-behaved function, it is square-integrable, so $\int |\chi|^2 dx$ is positive and finite. Hence $\lambda = \lambda^*$: the eigenvalue λ is real. This is why measurement results are real.

Property 2 — eigenfunctions for distinct eigenvalues are orthogonal. Suppose a Hermitian operator $\hat{A}$ has two different eigenfunctions, χ and ξ, with distinct eigenvalues λ and η respectively, $\hat{A}\chi = \lambda\chi$ and $\hat{A}\xi = \eta\xi$. From the definition of a Hermitian operator, with $\psi = \chi$ and $\phi = \xi$,

$$\int \xi^* \hat{A} \chi \, dx - \int \chi (\hat{A}\xi)^* \, dx = 0$$

Putting in the eigenvalue equations,

$$\lambda \int \xi^* \chi \, dx - \eta^* \int \chi \xi^* \, dx = 0$$

By Property 1 the eigenvalues are real, so $\eta^* = \eta$, and the two integrals are the same. Subtracting,

$$(\lambda - \eta) \int \xi^* \chi \, dx = 0$$

Since $\lambda \neq \eta$ by assumption, the integral itself must be zero:

$$\int \xi^* \chi \, dx = 0 \qquad (\lambda \neq \eta)$$

So the eigenfunctions which correspond to distinct eigenvalues of a Hermitian operator are orthogonal.

Property 3 — completeness. The eigenfunctions of a Hermitian operator form a complete orthonormal set: any well-behaved function can be expanded as $\psi = \sum_n c_n \psi_n$ with $c_n = \int \psi_n^* \psi \, d\tau$. To every observable there corresponds a Hermitian operator whose eigenvectors form a complete orthonormal basis.

Algebra of Hermitian operators. The sum of two Hermitian operators is Hermitian. The product $\hat{A}\hat{B}$ is Hermitian if and only if $\hat{A}$ and $\hat{B}$ commute. The commutator of two Hermitian operators is anti-Hermitian, while $i[\hat{A}, \hat{B}]$ is Hermitian.

10. The four postulates of quantum mechanics, and zero-point energy

Postulate 1 — the state. The state of a quantum mechanical system is completely specified by a function ψ, the wave function, normalised, $\int |\psi|^2 d\tau = 1$. $|\psi|^2$ is the probability density.

Postulate 2 — observables. To every observable in classical mechanics there corresponds a linear, Hermitian operator in quantum mechanics. The only possible results of measuring it are its eigenvalues.

Postulate 3 — measurement. If $\psi = \sum_n c_n \psi_n$ in the eigenfunctions of $\hat{A}$, a measurement of A gives a_n with probability $|c_n|^2$, and after the measurement the system is in the state ψ_n. The expectation value is $\langle A \rangle = \int \psi^* \hat{A} \psi \, d\tau$.

Postulate 4 — time evolution. ψ evolves in time by the time-dependent Schrödinger equation, $i\hbar \, \partial\psi/\partial t = \hat{H}\psi$.

Zero-point energy. According to kinetic theory, at absolute zero the atoms would stop oscillating and remain fixed. But according to the uncertainty relation, both position and momentum cannot be specified at the same time with complete accuracy, so the atomic oscillators, even at absolute zero, retain a certain amount of oscillatory motion, enough to obey the uncertainty relation. The energy possessed by the atomic oscillator at absolute zero is termed as zero-point energy.

Consider a particle of mass m, confined to a small one-dimensional region of size a. By the uncertainty relation the minimum uncertainty in its momentum is $\Delta p_x \sim \hbar/2a$. Since the momentum has a spread of at least $\hbar/2a$, the particle cannot be at rest: it must have a finite energy corresponding to a momentum of $\hbar/2a$. A quantum particle confined to a region must have an energy at least equal to

$$E_0 = \frac{\hbar^2}{8ma^2}$$

This energy is called the zero-point energy. For a particle in a one-dimensional box of length a, $E_1 = h^2/8ma^2$; for a harmonic oscillator, $E_0 = \frac{1}{2}\hbar\omega$. A quantum particle can never sit at rest at the bottom of a potential well.

11. Worked checks

(a) Which wave functions are admissible? $\psi = A \sec x$ and $\psi = A \tan x$ blow up where $\cos x = 0$ — not finite, reject. $\psi = Ae^{x^2}$ blows up as $|x| \to \infty$ and cannot be normalised — reject. $\psi = Ae^{-x^2}$ is finite, single-valued, continuous and square-integrable — admissible (it is a Gaussian). (Calicut PHY5 B07, Q14, 5 marks.)

(b) Commutator check. $[\hat{x}, \hat{p}_x^2] = 2i\hbar\hat{p}_x$ (Section 5) and $[\hat{p}_x, \hat{x}^2] = -2i\hbar\hat{x}$. Watch the sign: $[\hat{p}_x, \hat{x}] = -i\hbar$.

(c) Is $\hat{p}_x$ Hermitian? Yes (Section 9). Is d/dx Hermitian? No — it is anti-Hermitian: $(d/dx)^\dagger = -d/dx$. Is $\hat{x}\hat{p}_x$ Hermitian? No, because $\hat{x}$ and $\hat{p}_x$ do not commute; $(\hat{x}\hat{p}_x)^\dagger = \hat{p}_x\hat{x} \neq \hat{x}\hat{p}_x$.

(d) Simultaneous measurement. Energy and momentum of a free particle can be measured together ($[\hat{H}, \hat{p}_x] = 0$). Position and momentum cannot ($[\hat{x}, \hat{p}_x] = i\hbar$). This is the whole content of "compatible vs incompatible" in one line each.

12. Revision formula box

$\rho = |\psi|^2, \ \int |\psi|^2 d\tau = 1$

$i\hbar \dfrac{\partial \psi}{\partial t} = \left[-\dfrac{\hbar^2}{2m}\nabla^2 + V \right] \psi, \ \hat{H}\phi = E\phi$

$\mathbf{J} = \dfrac{\hbar}{2mi}(\psi^* \nabla \psi - \psi \nabla \psi^*), \ \dfrac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$

$\hat{p}_x = -i\hbar \dfrac{d}{dx}, \ \hat{T} = -\dfrac{\hbar^2}{2m}\dfrac{d^2}{dx^2}, \ \hat{H} = \hat{T} + V$

$[\hat{x}, \hat{p}_x] = i\hbar, \ [\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$

$\hat{A}\psi_n = a_n \psi_n, \ [\hat{A}, \hat{B}] = 0 \Rightarrow$ simultaneous eigenfunctions

Hermitian: eigenvalues real; eigenfunctions for $a_m \neq a_n$ orthogonal

$\langle A \rangle = \int \psi^* \hat{A} \psi \, d\tau, \ \Delta x \, \Delta p_x \geq \hbar/2$

Questions

Short form first (2–3 marks), then long form (5–10 marks). AU = Assam University exact question from the AU bank; non-AU and standard questions are marked as such.

Short-form questions

S1. [AU] Write four important postulates of quantum mechanics. — CHMHCC-502T, 5th Sem, 2020, CBCS, Sec A.

Answer in brief: Section 10 — (1) state described by normalised ψ; (2) observables are Hermitian operators, results are eigenvalues; (3) measurement gives a_n with probability $|c_n|^2$, state collapses to ψ_n; (4) time evolution by the TDSE.

S2. [AU] Write any two postulates of quantum mechanics. — CHMHCC-502T, 5th Sem, 2021, CBCS, Sec A.

Answer in brief: Any two from Section 10, e.g. Postulates 1 and 2, stated in full sentences.

S3. [AU] What do you mean by the term "zero point energy"? — CHMHCC-502T, 5th Sem, 2020, CBCS, Sec A.

Answer in brief: The lowest, non-zero energy of a quantum system (Section 10): $E_1 = h^2/8ma^2$ for a particle in a box, $E_0 = \frac{1}{2}\hbar\omega$ for an oscillator. It follows from the uncertainty principle; the particle can never be at rest.

S4. [AU] Prove that the linear combination of well-behaved wave functions is also well-behaved. — CHMHCC-502T, 5th Sem, 2023, CBCS, Sec A.

Answer in brief: Section 1 — a sum of finite, single-valued, continuous, square-integrable functions keeps each of those properties; normalise the sum afterwards.

S5. [non-AU] What is meant by eigenfunction and eigenvalue? — Univ. of Calicut, PHY5 B07, Nov 2021, Q9 (2 marks).

Answer in brief: Section 6 — $\hat{A}\psi_n = a_n\psi_n$; a_n is the eigenvalue, ψ_n the eigenfunction. A measurement of A can only give an eigenvalue.

S6. [standard] Define a linear operator and a Hermitian operator. Is d/dx Hermitian? Is $\hat{p}_x$ Hermitian?

Answer in brief: Sections 4 and 9 — linear: $\hat{A}(c_1\psi_1 + c_2\psi_2) = c_1\hat{A}\psi_1 + c_2\hat{A}\psi_2$. Hermitian: $\hat{A}^\dagger = \hat{A}$. d/dx is anti-Hermitian; $\hat{p}_x = -i\hbar\, d/dx$ is Hermitian.

Long-form questions

L1. [AU, paraphrased] If $[\hat{A}, \hat{B}] = \ldots$, then show that . . . are Hermitian operators, and find . . . such that . . . have real eigenvalues. — CHMHCC-502T, 5th Sem, 2021, CBCS, Sec A. (The bank entry is paraphrased — the original wording is partly garbled in the extracted paper text, so learn the general proof, not this wording.)

What to write: Section 9, Properties 1 and 2 — prove eigenvalues of a Hermitian operator are real, prove eigenfunctions for distinct eigenvalues are orthogonal, and use the commutator identities of Section 4–5 (e.g. $i[\hat{A}, \hat{B}]$ is Hermitian when $\hat{A}, \hat{B}$ are Hermitian).

L2. [standard] Derive the probability current density $\mathbf{J}$ and the equation of continuity, and show conservation of probability.

What to write: Section 3 in full — TDSE and its conjugate, multiply and subtract, identify $\rho = \psi^*\psi$ and $\mathbf{J} = \frac{\hbar}{2mi}(\psi^*\nabla\psi - \psi\nabla\psi^*)$, get $\partial\rho/\partial t + \nabla\cdot\mathbf{J} = 0$, integrate over all space to get $d/dt\int|\psi|^2 d\tau = 0$.

L3. [standard] Prove that the eigenvalues of a Hermitian operator are real, and that its eigenfunctions corresponding to different eigenvalues are orthogonal. Distinguish compatible and incompatible observables, with examples.

What to write: Section 9, Properties 1–2 (both proofs in full), then Section 7 with the table: $[\hat{x},\hat{p}_x] = i\hbar$ (incompatible) vs $[\hat{x},\hat{p}_y] = 0$ (compatible).

L4. [non-AU] Which of the following wave functions cannot be solutions of Schrödinger's equation for all x? Why not? (a) $\psi = A\sec x$ (b) $\psi = A\tan x$ (c) $\psi = Ae^{x^2}$ (d) $\psi = Ae^{-x^2}$. — Univ. of Calicut, PHY5 B07, Nov 2021, Q14 (5 marks).

What to write: Section 11(a) — (a) and (b) diverge where $\cos x = 0$; (c) diverges as $|x|\to\infty$ and is not normalisable; (d) is admissible. State the well-behaved conditions from Section 1 for each.

L5. [standard] Derive $[\hat{x},\hat{p}_x] = i\hbar$ and hence show $[\hat{x},\hat{p}_x^2] = 2i\hbar\hat{p}_x$. State and explain the condition for two operators to have simultaneous eigenfunctions.

What to write: Section 5 for the derivation (act on a general ψ), the product identity $[\hat{A},\hat{B}\hat{C}] = [\hat{A},\hat{B}]\hat{C} + \hat{B}[\hat{A},\hat{C}]$, then Section 6 for simultaneous eigenfunctions: $[\hat{A},\hat{B}] = 0$ is the condition.

Sources

These notes are based on the following published lecture notes, in the order the user asked for: Indian colleges and universities first, then IIT / NPTEL, then open-learning (IGNOU), then international. The AU questions are from the Assam University paper bank (see the Questions section above), not from these sources.

Indian colleges and universities

- Institute of Mathematical Sciences (IMSc), Chennai — B. Sathiapalan, Quantum Mechanics course notes (wave function, Born interpretation, probability current). https://www.imsc.res.in/~bala/QM.pdf
- IISER Bhopal — CHM439 Lecture 22, Continuity equation (probability current density, continuity equation, conservation of probability). http://helios.iiserb.ac.in/Courses/CHM439/Lectures/Lecture22.pdf
- Aligarh Muslim University, Dept. of Physics — Quantum Mechanics Unit 1, B.Sc. (Hons.) V Sem, PHYSICS-C VIII (Hermitian operators, eigenvalues, commutator algebra, completeness). https://api.amu.ac.in//storage/file/study_material/674567246_10056724.pdf
- Sathyabama Institute of Science and Technology, Chennai — SPH1314 Unit I, Introduction to Quantum Mechanics (postulates, operators, zero-point energy, review questions). https://sist.sathyabama.ac.in/sist_coursematerial/uploads/SPH1314.pdf
- NIT Srinagar — Dr Prince A. Ganai, Introduction of Quantum Mechanics (five postulates, expectation values, Hamiltonian operator). https://www.nitsri.ac.in/Department/PHYSICS/lecture_08_Introduction_QM_.pdf
- IFAS Publications, Pune — Quantum Chemistry, Ch. 1 (linear operators, commutator properties, Hermitian operators, eigenvalues, postulates, TISE). https://ifaslive-preprod-s3.s3.ap-south-1.amazonaws.com/courses/a280c718-2194-42f9-8ede-8f36528c9e10.pdf
- JECRC Foundation, Jaipur — Quantum Mechanics – II, Engineering Physics notes (TDSE and TISE derivations). https://jecrcfoundation.com/wp-content/uploads/notes/btech/Computer%20Science%20Engineering/2nd%20Semester/Engineering%20Physics/Quantum%20Mechanics%20-%20II.pdf

IIT / NPTEL

- IIT Madras (NPTEL) — Prof. V. Balakrishnan, Quantum Physics, Lecture 14 (probability density, probability current derivation, continuity equation). http://kcl.digimat.in/nptel/courses/video/122106034/lec14.pdf
- IIT Madras (NPTEL) — Prof. S. Lakshmi Bala, Quantum Mechanics I, Module 35: Algebra of Operators, Lecture 35 (Hermitian operators, real eigenvalues, orthogonality, commutators, uncertainty relation). https://archive.nptel.ac.in/content/storage2/courses/115105083/lec-35.pdf
- IIT Kanpur — CHM102, Lecture 2: Operators, Eigenfunctions and the Schrödinger Equation. http://home.iitk.ac.in/~madhavr/CHM102/Physical/Lec2.pdf

- IIT Delhi (NPTEL) — Prof. Ajoy Ghatak, Basic Quantum Mechanics, Module 7 Lecture 2 (linear operators, adjoint operators, [x, p] commutator, eigenvalues). http://kcl.digimat.in/nptel/courses/video/115102023/lec25.pdf
- IIT Madras (NPTEL) — Prof. Mangala Sunder Krishnan, Chemistry I / Introduction to Quantum Chemistry, Lecture 12 (expectation value as a postulate, worked examples). http://kcl.digimat.in/nptel/courses/video/104106083/lec12.pdf

Open learning (IGNOU)

- IGNOU (eGyanKosh) — BPHET-141 Elements of Modern Physics, Block 2: Introduction to Quantum Mechanics (wave function, Schrödinger equation, operators, commutators, Hermitian operators, expectation values, postulates, zero-point energy). https://www.egyankosh.ac.in/bitstream/123456789/76289/1/Block-2.pdf
- IGNOU (eGyanKosh) — MPH-004 Quantum Mechanics-I, Block 3, Unit 11: Observables and Quantum Dynamics (compatible and incompatible observables, simultaneous eigenvectors, sequential measurement). https://egyankosh.ac.in/bitstream/123456789/98955/1/Unit-11.pdf

International (used only for the written form of the probability-current step)

- MIT OpenCourseWare — Prof. Barton Zwiebach, Quantum Physics I (8.04), Lecture Note 6 (probability current in 1D and 3D, current conservation). https://github.com/manjunath5496/Physics-Tutorial/raw/refs/heads/master/Lectures/MIT8_04S16_LecNotes6.pdf (mirror of the MIT OCW notes)